Research problem

Are reported detector improvements on NEU-DET and GC10-DET larger than split, seed, and duplicate-leakage noise?

A Literature evidence audit

131 papers, 182 evaluations

Systematic PRISMA search
433 Europe PMC records;
131 PMC journal studies

17-field protocol audit
F01–F17 verbatim coding
with Wilson 95% CIs

Claimed-gain extraction
174 ΔmAP@0.5 comparisons
(median +3.80 pp)

Reporting compliance gaps
0% manifests; 89% no SD

B Controlled benchmark

170 clean cloud runs

Duplicate & coil grouping
223 SE(2) edges (NEU-DET);
467 coils (GC10-DET)

10-seed detector baselines
5 detectors × 10 seeds on
zero-leakage splits (100 runs)

Split ANOVA & modules
10 grouped splits × 3 seeds +
10-seed CBAM/EMA/BiFPN

Split & seed noise floors
8.87 pp split; 2.6–6.6 pp seed

C Counterfactual leakage

127 CF runs + escape audit

K=10 partner replacement
hold test images fixed;
swap only 31 train partners

5-dose leakage scaling
0% to 100% partner doses
(matched K=0..2 × 3 seeds)

Precision & escape metrics
Pooled Micro-AP, Recall &
matched-FP escape rate

Causal leakage inflation
+4.08 pp mAP; −3.0 pp escape

Evidence synthesis

38.5% of claims ≤ 3.0 pp;
10-seed modules null (+0.01–0.33 pp)

Inspection practice

fixed-FP escape drops −21% to −25%
under cross-split leakage

Open benchmark kit

grouped manifests, SHA-256 hashes,
297-run logs & audit code